RAM PRASATH K

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PROFESSIONAL SUMMARY

Machine Learning Engineer with **2+ years** building production AI systems across agentic workflows, **LLM/VLM serving, retrieval, and backend infrastructure**. Experienced with MCP, multi-agent systems, RAG, evaluation, **vLLM, FastAPI, and Docker,** with a focus on making AI systems reliable and measurable in production.

EXPERIENCE

UBISOFT INDIA STUDIOS,R&D — Pune, India

JR. R&D ENGINEER — *October 2025 – Present*

Game AI & Quality Control division

- Investigated training issues in **NVIDIA NitroGen (Gr00t), a Vision-Language-Action model**, by tracing problems across the data pipeline, optimizer, and model architecture, improving training reliability for internal Game AI workflows.
- Built **EagleEye Snap inside Ubisoft's C++ Anvil engine** to detect triangle-level mesh clipping, integrating a **LoRA fine-tuned Qwen-VL** classifier to distinguish intentional versus unintended geometry intersections.
- Designed a **hybrid RAG system over Perforce changelist history** combining BM25, vector retrieval, and reranking to **accelerate regression root-cause analysis** across game builds.

Core stack: PyTorch · Diffusion Transformers · C++ · C# · Anvil · P4V · Jira

NINESTARS INFORMATION TECHNOLOGIES,R&D — Bengaluru, India

DATA SCIENCE ENGINEER — *July 2024 – Sept 2025*

R&D division focused on production AI systems for OCR

- Integrated **YOLO + RT-DETR** for layout-aware annotation acceleration via **Flask backend**, boosting **labeling throughput 4×, reducing manual review by 20%,** and lifting entity extraction precision by 15%.
- Built backend inference services using **FastAPI, PostgreSQL, and gRPC-based communication layers** to support scalable document understanding workflows and low-latency model serving.
- **Optimized VLM inference with vLLM (W8A16/W4A16 quantization)** on RTX 4090 infrastructure, profiling TTFT, decode throughput, request concurrency, and GPU memory to hold stable low-latency serving in production.
- Designed and implemented a **multi-agent automation platform using Google ADK and MCP**, serving a self-hosted **GPT-OSS-20B** model through **vLLM** and instrumenting tool-call evaluation with Langfuse and achieving **92% tool-call success** across enterprise workflows.

Core stack: Python · PyTorch · Hugging Face · vLLM · AWS · FastAPI · Docker · WebSockets

BOSCH GLOBAL SOFTWARE TECHNOLOGIES,HMI — Bengaluru, India

RESEARCH ENGINEER INTERN — *Aug 2023 – July 2024*

- Developed a **ResNet-SE**–based automotive voice authentication system with anti-spoofing capabilities, achieving **95% accuracy and 4% Equal Error Rate (EER)** under real-world noisy driving conditions. Trained multilingual speaker recognition models using **VoxCeleb and AI4Bharat IndicSuperb datasets.**
- Optimized and deployed inference pipelines using **ONNX and TensorFlow Lite on NXP edge hardware**(Android Target), collaborating with HMI and embedded teams to satisfy production latency and memory constraints for in-vehicle voice systems.
- Developed an experimental speaker diarization pipeline for multi-speaker automotive audio using **ECAPA-TDNN embeddings and clustering techniques**.

Core stack: PyTorch · TensorFlow Lite · ONNX · NXP · Android · VoxCeleb · IndicSuperb

SKILLS

Languages: Python, C++, C#, SQL

ML & Generative AI: PyTorch, HF Transformers, fine-tuning (SFT, LoRA), RAG, multi-agent systems (MCP, Google ADK)

Inference & Serving: vLLM, quantization (AWQ, GPTQ, SmoothQuant, FP8), KV-cache optimization, continuous batching, ONNX, latency/throughput profiling

GPU & Edge Platforms: NVIDIA RTX 4090, AMD Instinct MI300X, NXP edge hardware (Android)

Backend & Infrastructure: FastAPI, gRPC, PostgreSQL, Docker, AWS, Linux, CI/CD

Post-Training & Observability: DPO, PPO, GRPO (research), Weights & Biases, Langfuse, MLflow

PROJECTS

Differential-MoE: Differential Attention & Mixture-of-Experts for LLMs

- Trained four **379M-209M-active-parameter transformer models** from scratch on two Kaggle T4 GPUs to study differential attention and **top-2 Mixture-of-Experts.**
- In the experiments, the MoE model improved held-out test **NLL by 0.129 nats** at the same active compute budget.

IssueFix-RL, RL Post-Training for Automated Code Repair, *(in progress)*

- Building a framework to compare **SFT, PPO, GRPO, and DR-GRPO** for automated GitHub issue resolution using **Qwen2.5-0.5B**.
- Exploring **reasoning-oriented training for code repair**, investigating whether explicit intermediate reasoning helps models better understand an issue, plan and fix.
- Designed **execution-based rewards using unit-test results, linting, and compilation outcomes** to measure whether generated fixes actually work.

TripleQuant-VLM - KV Cache & Weight Quantization Research

- Unified benchmark across **AWQ, GPTQ, SmoothQuant, and FP8** on two runtimes. The same W4A16 checkpoint runs **4.1 tok/s on Hugging Face eager** versus **57.9 tok/s under vLLM Marlin kernels** - a 14x gap from runtime choice alone.
- Implemented a KV-cache compression method using **orthogonal rotation** and **codebooks**, increasing usable context from **4K to 16K tokens on a 12GB RTX 3060.**

ACHIEVEMENTS

Mistral AI Worldwide Hackathon 2026 - Winner, #1 (Online Track):

- Built Mistral Raid, a dungeon crawler where an AI-driven boss reacts to player voice and gameplay using Mistral agents, **Voxtral STT, and real-time combat telemetry.**
- Built a telemetry pipeline that streamed game state such as player position and health to the agents so the boss could adapt its behavior during combat.

AMD, Unsloth, PyTorch Synthetic Data AI Agents Challenge (Nov 2025):

- Designed and fine-tuned QA AI agents using synthetic data pipelines and Synthetic-Data-Kit on AMD Instinct **MI300X GPUs,** evaluated **SFT, LoRA, QLoRA, and DPO fine-tuning workflows on next-gen GPU architecture.**

Appearance-Trajectory Network for Video Anomaly Detection System, ICCCNT 2024, IEEE, IIT Mandi (July 2024):

- Memory-augmented CNN for unsupervised video surveillance achieving 82% AUC (appearance) and 84.9% AUC (skeleton), with integrated video captioning for explainability.

Simultaneous Voice Authentication and Decision-Making Based on Intent & Entity Classification, Patent No. 202441102999 (filed Dec 24, 2024):

- System combining voice authentication with real-time intent and entity classification for secure, context-aware voice-driven decision automation.

EDUCATION

AMRITA VISHWA VIDYAPEETHAM, School of Artificial Intelligence, Coimbatore, India
Master of Technology in Data Science, CGPA 8.5/10 *July 2022- Jun,2024*

INDIAN INSTITUTE OF TECHNOLOGY–MADRAS (IIT–MADRAS) 🔗 Chennai, India
Bachelor of Science in Data Science and Applications ***(Online)*** *Dec 2020 - Jan 2024*

SASTRA DEEMED UNIVERSITY Thanjavur, India
Bachelor of Technology in Mechanical Engineering, CGPA 8/10 *Apr 2018 - Jun 2022*